

Smart Portable Real Time Detection System for Microplastics In Water Bodies

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Abstract— Microplastic pollution in water has emerged as a major environmental problem because of its toxic impacts on the environment and human health. The detection of microplastic particles in water using conventional laboratory methods such as FTIR and Raman spectroscopy is a costly, time-consuming, and highly technical process. This research work proposes the development of a Low-Cost Portable Microplastic Detector (100-300 μm) using optical sensing and image processing techniques. The proposed system combines hardware components such as an ESP32 microcontroller, camera module, white LED illumination source, laser diode, photodiode sensor, and flow cell chamber to detect and analyze microplastic particles in water samples. The detected images and scattering patterns are analyzed using software tools such as Python, image processing algorithms, and machine learning algorithms to classify particles according to size and optical properties. The proposed system will enable the development of a portable, low-cost, and real-time monitoring device for environmental water quality analysis.

Keywords—Microplastics detection; Image Processing; Optical Sensing; Machine Learning.

While these approaches are accurate, they are quite costly and complicated, requiring specialized equipment and expertise. This makes them unsuitable for field testing or environmental studies that need to be done on a budget. There is, therefore, a need for a system that can detect microplastic particles in water samples with a reasonable level of accuracy and without breaking the bank.

The proposed project involves the design and development of a Low-Cost Portable Microplastic Detector (100-300 μm) based on optical sensing and image analysis. The proposed system employs a combination of hardware components such as an ESP32 microcontroller, camera module, white LED illumination source, laser diode, and a photodiode sensor along with a transparent flow cell chamber to analyze the microplastic particles suspended in the water sample. The LED illumination source is used for uniform illumination of the image, while the laser and photodiode sensor are used to detect the scattering of light caused by the particles suspended in the chamber.

I. INTRODUCTION

Microplastics refer to plastic particles that are less than 5mm in size and are found in large quantities in oceans, rivers, lakes, and even drinking water. These microplastics are formed when larger plastic materials are broken down or through the use of microbeads in cosmetics and industrial materials. Because of their small size, microplastics can easily find their way into aquatic environments and be ingested by marine animals, eventually making their way into the human food chain. The traditional approaches that can be used to detect microplastics include laboratory analysis using techniques such as Fourier Transform Infrared Spectroscopy (FTIR), Raman spectroscopy, and scanning electron microscopy.

II. LITERATURE REVIEW

Microplastics are plastic particles that are less than 5mm in size and are found in large quantities in oceans, rivers, and freshwater bodies. The detection and measurement of microplastics in water are difficult because of their size, irregular shapes, and transparency.

Conventional techniques like microscopy, Fourier Transform Infrared Spectroscopy (FTIR), and Raman spectroscopy are precise but require costly equipment and sophisticated laboratory settings.

Hence, there is a need for automated and real-time detection systems using computer vision and machine learning algorithms. The proposed system consists of a camera-based image acquisition system to acquire images of water samples containing microplastics. The acquired images are used to train an object detection model. The researchers have used the YOLOv5 (You Only Look Once version 5) object detection model, which is famous for its fast and precise real-time detection mechanism. The model is trained using manually labeled images of microplastic particles to identify various shapes and colors of microplastics. This model detects the particles in each consecutive frame of the video and helps the system to trace the movement of each particle and prevents the system from counting the same particle twice.

The prototype designed in the study employs a camera sensor for image acquisition and analyzes the video stream using the trained YOLOv5 model. The prototype is capable of detecting microplastic particles between 1mm and 5mm in size and is functional at water flow velocities of up to 34 cm/s. The experimental results indicate that the prototype is able to successfully detect, track, and count microplastics in real-time, thus proving the effectiveness of employing AI-based computer vision for environmental monitoring. The study concludes that deep learning-based optical detection systems have the potential to greatly enhance the speed and automation of microplastic monitoring compared to conventional laboratory methods. However, the authors also state that there is a need for further enhancement in the system for the detection of smaller particles and for its practical implementation in environmental conditions.

III. METHODOLOGY

1. **Sample Preparation and Flow Cell Setup:** The initial process of the system entails the collection of water samples that may contain microplastic particles. The collected water is then subjected to a transparent flow cell chamber constructed from acrylic or glass materials. The flow cell chamber enables the creation of a controlled path through which the sample passes. The chamber is also designed to ensure that the particles move into the optical sensing region in a stable and uniform manner.
2. **Illumination and Optical Detection:** To facilitate easy observation of the particles, a white LED light source is employed to illuminate the sample uniformly inside the flow cell chamber. The light source assists the camera module in taking clear images of the particles suspended in the water. Besides the use of the LED light source, a laser diode is employed to produce a focused beam of light that passes through the flowing sample. When microplastic particles interact with the laser beam, the light scatters in various directions based on the size and shape of the particles. A photodiode sensor is placed at a particular angle to measure the scattered light from the particles. The measured optical signal depends on the concentration and size of particles suspended in the sample.
3. **Signal Amplification and Data Acquisition:** The output signal from the photodiode is normally very weak and needs amplification before it can be processed. An operational amplifier (LM358) is used to amplify the signal from the photodiode. The amplified signal is then

transmitted to the ESP32 microcontroller, which interprets the analog voltage using its in-built analog-to-digital converter. The signal is an indication of the intensity of the scattered light, which is an indication of the presence of particles in the sample.

4. **Image Capture and Processing:** A camera module is connected to the ESP32 to capture images of the particles being illuminated as they pass through the chamber. The images are then transmitted to a computer system where they are processed using Python-based image processing algorithms. The algorithms involve grayscale conversion, noise removal, thresholding, and contour detection to separate the particles from the background. After the particles have been detected in the images, their size, shape, and number are determined. Particles that fall within the 100-300 μm range are then chosen for analysis.
5. **Machine Learning-Based Classification:** To increase the detection accuracy, a simplified machine learning algorithm like Decision Tree is employed. The trained algorithm uses the extracted features from the processed images, including particle size, intensity, and scattering properties. The trained algorithm then identifies the detected particles as microplastics or other debris.
6. **Result Display and Analysis:** Finally, the processed data is shown on a serial monitor or graphical user interface. The system provides data on the number of particles, size distribution, and relative concentration of microplastics in the sample. This allows for rapid assessment of water quality using a portable and affordable detection system.

A. System Overview

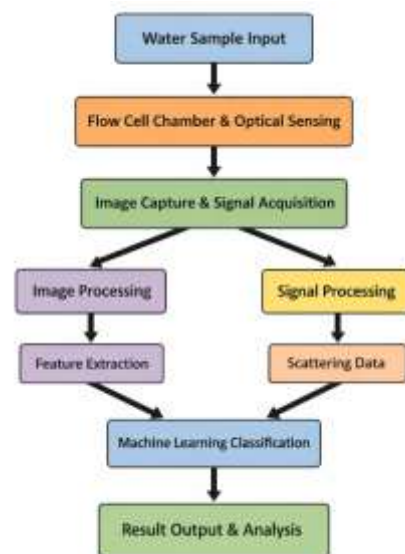


Fig 1 SYSTEM OVERVIEW

The proposed system uses optical sensing and image processing to identify microplastic particles in water. A water sample is injected into a transparent flow cell chamber where a white LED shines on the particles.

A laser diode transmits light through the water, and scattered light from particles is measured by a photodiode sensor. The data is amplified and interpreted by an ESP32 microcontroller.

Simultaneously, a camera module takes pictures of the particles. The images are then analyzed by Python image processing and machine learning software to detect particles in the 100-300 μm size range and determine their concentration in the water sample.

B. Hardware Setup

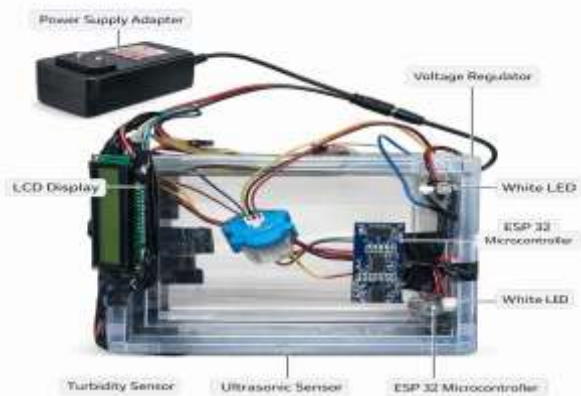


Fig 2 HARDWARE SETUP

Prototype developed is compact and integrated smart monitoring system comprising of an ESP32 microcontroller. System development was done in such a way that it can be viewed from outside and monitored at any time.

Power supply to the whole system is provided using an external adapter. Voltage regulator ensures stable voltage distribution among the different parts of the system. As a result, this improves the efficiency of the system and prevents any damage which might have been caused by voltage fluctuations.

The turbidity sensor used in the system helps in measuring the amount of clarity or dirtiness of the liquid inside the chamber. This makes the system ideal for use in the process of water quality measurement. In addition to the turbidity sensor, the ultrasonic sensor is used in determining the liquid level in order to ensure continuous monitoring and avoid overflow or dry run situations.

The presence of the LDR and the white LED shows that illumination is used in some aspects to control sensing process. Lastly, the LCD serves as a screen where real-time sensor measurements are displayed. ESP32 microcontroller serves as the main processor in the system.

C. Features of proposed system

1. **Low-Cost Design:** The system is designed using low-cost components like ESP32, LED, photodiode, and camera module.
2. **Portable Device:** The system is designed in a compact manner that makes it portable and can be used for on-site water quality analysis.
3. **Optical Sensing Technique:** The system uses LED illumination and laser-based light scattering to detect the particles in the water sample.
4. **Camera-Based Detection:** The system uses a camera module to take images of the particles flowing through the chamber.
5. **Real-Time Monitoring:** The system can analyze the particles continuously as the water flows through the sensing chamber.
6. **Image Processing Integration:** The system uses image processing algorithms to analyze the particles based on size, shape, and count.
7. **Machine Learning Classification:** The system uses machine learning models to classify the particles in the 100-300 μm microplastic range.
8. **Dual Detection Method:** The system uses optical scattering sensing and image analysis for improved detection accuracy.
9. **Automatic Data Processing:** The system automatically processes the sensor and image data to produce results.
10. **Environmental Monitoring Application:** The system can be used for monitoring microplastic contamination in rivers, lakes, and other water bodies.
11. **Regenerative Braking System :** When braking, the system harvests kinetic energy and converts it to electrical energy. This energy is stored in the secondary battery through a charge controller, enhancing overall efficiency and battery life.

IV. RESULTS

Machine learning-based efficiency prediction and analytical estimation based on real-time data acquisition and calculations.

A. Machine learning-based prediction

The system shows efficiency in detection of the microplastics using the optical sensor in terms of varying the light intensity. Light emitted from the white LED is passed across the liquid samples where it encounters some variations when particles block its path. LDR receives the varying light intensities and processes the data using ESP32 to detect any particle presence.

Analog inputs are converted to digital values and further filtered using thresholding techniques, and only meaningful light variations are considered to ensure accurate detection. Additionally, turbidity levels and water height are considered to increase reliability.

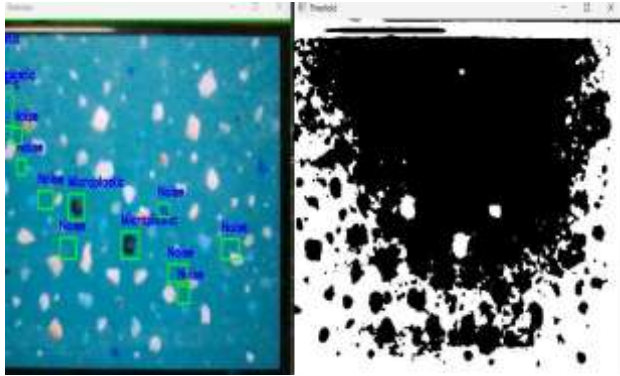


Fig 3 DETECTING MICROPLASTICS USING MACHINE LEARNING

From the above parameters, features that include varying light intensities, turbidity, and water volumes can be extracted to determine particles' presence and distribution. Based on the parameters, the system classifies contaminated water into different levels such as low, moderate, and high.

In the end, output is presented on the LCD screen in real-time showing the estimated number of particles, turbidity levels, and height of water. The system is efficient in detecting particles within the 100-300 μm range approximately.

B. Sem Analysis

The presence of heterogeneous particulate material within the tested water samples can be observed under different magnifications through the images taken using the Scanning Electron Microscope (SEM) (250 $\times$, 1.00 k $\times$, and 4.00 k $\times$). The 250 $\times$ magnification reveals that the surface is filled with irregularly distributed particles, demonstrating a high amount of suspension particles

Under 1.00 k $\times$ magnification, the specific shapes of particles can be seen, including angular fragments and rod-like particles. It should be stated that the elongated shape of the particles indicates fibrous materials that are made of synthetic textiles. The 4.00 k $\times$ image shows the rough texture of the particles on the surface with irregular edge structures, which are typical for plastic fragments but not natural minerals.

Thus, the SEM test proved the presence of microfibers and fragmented particles, demonstrating contamination. Consequently, it is necessary to mention that the test confirmed the presence of microplastic-like and other particulate materials in the water sample.

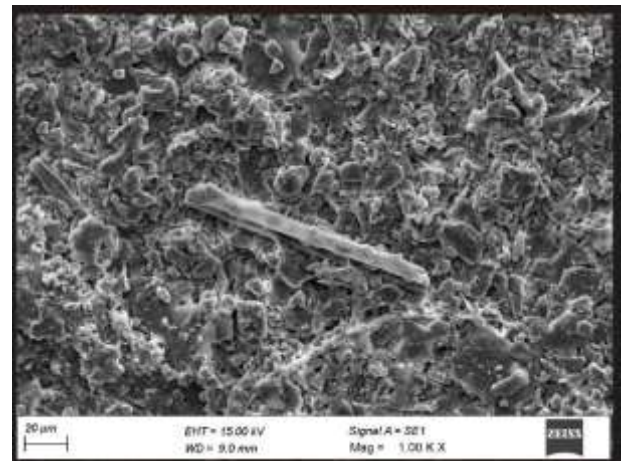
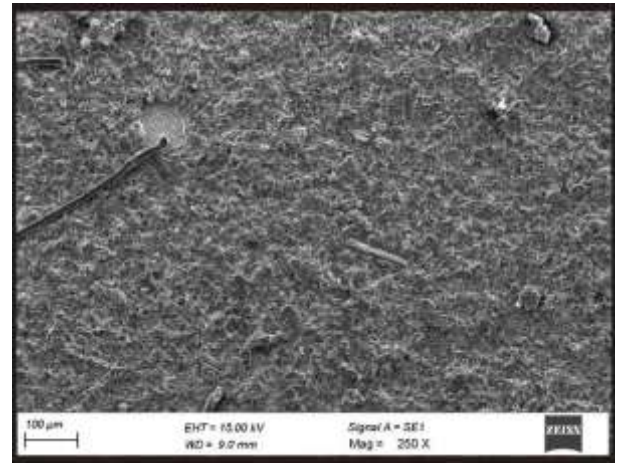


Fig 4 SEM ANALYSIS USING ELECTRON MICROSCOPE

Figure 4 depicts output obtained from the Scanning electron microscope (SEM) analysis of a water sample.

V. COMPARISION BETWEEN EXISTING SYSTEM AND PROPOSED SYSTEM

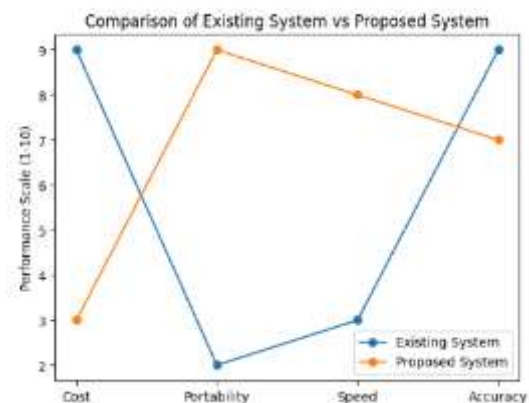


Fig 5 COMPARISION GRAPH

1. Cost: Present technologies such as FTIR and Raman spectroscopy are costly because of expensive laboratory equipment used in them, whereas in the proposed technology cost-effective devices including ESP32, sensors, and camera modules are used.

2. **Portability:** The conventional technologies are big and are restricted to laboratories only, whereas the proposed technology is small and portable and can be used in the field too.
3. **Speed:** The current methods involve lengthy sample preparations and analyses, but with the proposed technology, there is speedy and almost real-time detection.
4. **Accuracy:** The conventional technologies have high accuracy levels, whereas with the optical sensor and machine learning technology, the proposed method has an acceptable accuracy level.
5. **Usability:** The conventional methods need trained personnel, whereas in the proposed method usability is easy.
6. **Application:** The proposed method facilitates on-site testing rather than just laboratory testing.

VI. INSIGHTS AND IMPLICATIONS

The proposed system is useful because it offers a feasible and economical solution for the detection of microplastics in water resources. The proposed system uses optical sensing, image processing, and machine learning algorithms to achieve faster and more efficient monitoring than the traditional laboratory process. The proposed system is portable and can be applied in field settings, such as rivers, lakes, and wastewater sources. The proposed system can assist researchers and environmental organizations in gathering information about the concentration and size distribution of microplastics. This information is valuable for determining the levels of pollution and sources of plastic pollution. In general, the proposed system is useful for enhancing environmental monitoring and assisting in the protection of water quality and environments.

VII. FUTURE WORK

1. **Enhanced Accuracy of Detection:** Future studies can aim at enhancing the accuracy of detection by employing more sophisticated models of deep learning for better identification of microplastic particles.
2. **Detection of Smaller Particles:** The system can be developed to detect particles that are smaller than 100 μm , which cannot be easily detected by basic optical systems.
3. **IoT Integration:** The system can be integrated with IoT technology to allow remote monitoring and transmission of data in real time.
4. **Enhanced Imaging Systems:** More sophisticated cameras and optical systems can be employed to obtain clearer images of the particles.
5. **Field Testing of the System:** Future studies can aim at testing the system in various real-world settings such as rivers, lakes, and wastewater treatment plants.

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